

Externalities

▼ What are externalities

Refers to the costs or benefits experienced by a third party due to the consumption or production of a good or service

The third party has no control over the production or consumption of this good or service

Results in an inefficient market

▼ What is a positive consumption externality

Consumption creates benefits to a third party

Examples

- Education
- Flu vaccines

Results in social benefits greater than private benefits, demand curve is too far to the left which is an indication of *underconsumption*

Governments try to *raise consumption* to a socially correct level and reward consumers for socially good behaviour

▼ What is a negative consumption externality

Consumption creates costs to a third party

Examples

- Using phone while driving
- Smoking

Results in social benefits less than private benefits, demand curve is too far to the right indicating *overconsumption*

Governments try to *decrease consumption* to a socially correct level and discourage socially bad behaviour

▼ What is a positive production externality

Production reduces costs to third party

Example

- training workers
- renewable energy

Results in social costs lower than private costs, supply curve is too far left indicating underproduction

Governments try to increase production to a socially correct level and reward producers for socially good behaviour

▼ What is a negative production externality

Production imposes costs to a third party

Examples

- Using fossil fuels
- Loud music at festivals

Results in social costs greater than private costs, supply curve is too low indicating *overproduction*

Governments try to *reduce production* to a socially correct level and make polluters pay for socially bad behaviour

▼ What are government policies to fix externalities

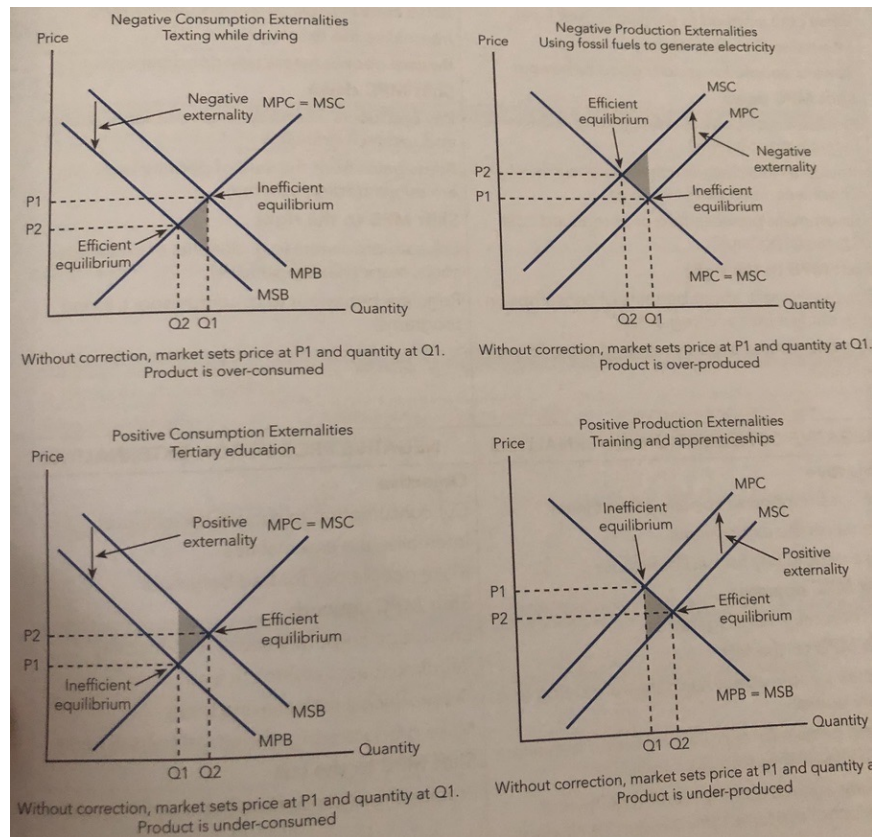
Tax

- Increase cost of production
- Increase marginal private cost
- Intended to reduce production to the optimum level and eliminate DWL

Subsidies

- Reduce cost of production
- Reduce marginal private cost
- Intended to increase production to the optimum level and eliminate DWL

▼ Evaluation of externality graphs



Above for negative production externality:

- $MSC > MPC$ because of the negative production externality, the cost to the community is greater than the cost to producers. The producers are producing too much and the cost of that production is too much because of the cost to society. This means the government needs to restrict production with something like a tax

Without externalities, MPB should = MSB to create the 'demand' curve. Same with marginal costs.

Above also:

1. Demand curves are marginal benefit curves
 1. Marginal private benefits = MPB
 2. Marginal social benefits = MSB
2. Supply curves are marginal cost curves
 1. Marginal private costs = MPC
 2. Marginal social costs = MSC

3. The inefficient equilibrium is at P_1 and Q_1 and is where private benefits (demand) and private costs (supply) are equal.
4. The impact of a consumption externality is shown as a move in the marginal benefits curve
 1. Right for a positive externality
 2. Left for a negative externality
5. The impact of a production externality is shown as a move in the marginal costs curve
 1. Up for negative externality
 2. Down for positive externality
6. The efficient or socially correct equilibrium is P_2 and Q_2 where social benefits (demand) and social costs (supply) are equal
7. DWL is the triangle

$MSB=MSC$ is optimal from society's POV as the opportunity cost of resources represents alternatives for all of society, we want our resources to be used as efficiently as possible. If production were inefficient, then it would be possible for some people to gain while imposing costs on others.

This is allocative efficiency where cost of production just meets benefits to consumers

Businesses will produce at the profit maximising quantity in a competitive market, where $Q_d=Q_s$. Society wants allocative efficiency where $MSB=MSC$. if there are no spill over costs or benefits, then $S=MSC$ and $D=MSB$ then:

Competitive markets achieve allocative efficiency

